



National University
Polyclinics

BRONCHIAL ASTHMA IN CHILDREN

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I. INTRODUCTION

Importance

Asthma is a chronic reversible airway disorder that is common in people of all ages. It can be severe and may be fatal. Asthma may present with cough, wheezing, and unexplained dyspnoea and chest tightness. Symptoms are often transient, may be persistent and tend to be worse at night or in the early mornings.

Management of asthma in children, particularly in children in the first five years of life, is often a challenge. Difficulties with diagnosis, efficacy and safety of drugs and drug delivery are common issues faced by the practitioner.

Definition of asthma is the same in children as in adults. A detailed medical history and clinical examination is mandatory. Additional tests like Pulmonary Function Tests (PFT) may be useful to support the diagnosis made or to monitor response to therapy for the more difficult patients.

II. PRESENTATION AND DIAGNOSIS

Presentation of Asthma in Children

- Definition of asthma is the same in children as in adults.
- Asthma should be considered if any of the following, cough, recurrent wheeze / breathing difficulty or chest tightness, is present. Symptoms often occur or worsen at night. Symptoms can also occur or worsen with exercise, or on exposure to various triggers, e.g. dust mite allergens.
- Asthma exacerbations in children are often triggered by respiratory viral and mycoplasma infections. However, it should also be noted that wheezing is commonly associated with viral respiratory illnesses especially in young children and this symptom alone without the presence of interval symptoms may not be due to asthma.
- The presence of atopy (such as asthma, eczema, allergic rhinitis) or a family history of atopy supports the diagnosis of asthma.
- Additional tests like Pulmonary Function Tests (PFT), exercise challenge test may be useful to support the diagnosis made or to monitor response to therapy for the more difficult patients.

Features suggesting a diagnosis of asthma in children:

Features	Characteristics Suggesting Asthma
Cough	Recurrent or persistent non-productive cough that may be worse in the middle of the night Cough occurring with exercise, laughing, crying or exposure to tobacco smoke (particularly in absence of respiratory infection)
Wheezing	Recurrent wheezing, including during sleep, or with triggers such as activity, laughing, crying or exposure to tobacco smoke or air pollution
Difficult or heavy breathing; or shortness of breath	Occurring with exercise, laughing, or crying
Activity limitation	Not running, playing, or laughing at the same intensity as other children, tires earlier during walks
Family or past personal history	Atopic dermatitis, allergic rhinitis, food allergy Asthma in first-degree relative(s)

History of variable respiratory symptoms:

- Reversible episodic cough or wheeze
- Symptoms occur variably over time and vary in intensity
- Symptoms are often worse at night or on waking
- Symptoms are often triggered by exercise, laughter, crying, allergens, cold air
- Symptoms often appear or worsen with viral infections

Supportive features

- Past personal history of atopic dermatitis, allergic rhinitis, food allergy or allergen sensitization
- Family history of asthma/ atopy

< 5 years old

≥ 5 years old

Spirometry

Reduced FEV₁ with reduced FEV₁/FVC ratio

Bronchodilator response:

Increase FEV₁ > 12% predicted after bronchodilator challenge

Positive exercise challenge test

Fall in FEV₁ of > 12% from pre-exercise values, or PEF > 15%

Commence on trial of asthma therapy for 8-12 weeks
Review diagnosis if response is poor
Consider referral to a Paediatrician for patients < 5 years of age

Watch out for Red Flags

Consider investigations in the presence of red flags:

1. Neonatal / early onset
2. Failure to thrive, LOW
3. Frequent vomiting / choking
4. Focal lung or cardiovascular signs
5. Continuous wheezing
6. No association of symptoms with typical triggers
7. Hypoxemia outside context of viral illness

Beware of alternative diagnosis:

1. Recurrent viral infections with wheezing
2. Chronic rhino-sinusitis
3. Gastro-oesophageal reflux
4. Bronchopulmonary dysplasia
5. Chronic lung disease of prematurity
6. Aspiration syndromes including foreign body aspiration / recurrent silent aspiration
7. Congenital malformations of lung
8. Congenital heart disease
9. Tuberculosis

Other Modes of Presentation

1. Cough variant asthma without wheezing
May be the group over-diagnosed as asthma, to rule out rhinitis and sinusitis
2. Recurrent viral wheezing in children aged 5 years or younger without atopy may not respond to asthma treatment
3. Exercise-induced bronchoconstriction

Investigations

Investigations are usually not necessary except in severe or atypical cases, and in patients who do not respond to therapy. Some investigations which may be considered are:

1. Chest X-Ray

To exclude foreign body, structural abnormalities, chronic chest infection or to exclude complications in severe acute episodes.

2. Pulmonary Function Tests (Spirometry)

Many children by 5 years old age are capable of performing spirometry if coached by experienced technician and with visual incentives. Children under 8 years of age are deemed to have completed the test if they have sustained expiratory effort for 3 seconds (as opposed to 6 seconds in adults) provided there is plateau of 1 second in the volume time graph. Children tend to empty their lungs faster.

3. Allergy Tests

Atopic status can be identified by skin prick testing or by measuring the level of specific immunoglobulin E (sIgE) in serum especially in patients with no clinical signs or family history of atopy, however, the presence of a positive skin prick test or positive sIgE, does not mean that the allergen is causing the symptoms - the relevance of allergen exposure and its relation to symptoms must be confirmed by the patient's history. Other allergy tests such as antigen specific IgG, IgG4, intradermal skin tests are not useful. Food allergy testing is also not useful for evaluation of asthma per se.

4. Airway Challenge Tests - Using Exercise or with Inhalation of Methacholine or Histamine

Methacholine or Histamine challenge tests are not routinely performed in clinical practice in children. Exercise challenge is however useful for evaluation of children with exercise-induced asthma symptoms.

III. MANAGEMENT

Asthma Control Goals and Outcomes

- No limitation of daily activities, including exercise
- No or minimal daytime symptoms
 - i. For children 5 years and younger: once or less/week
 - ii. For children 6-11 years: twice or less/week
- No nocturnal symptoms or awakening because of asthma
- No or minimal need for reliever treatment
 - i. For children 5 years and younger: once or less/week
 - ii. For children 6-11 years: twice or less/week
- No exacerbations
- Normal or near-normal lung function results

Initial Management After Diagnosis

- Good doctor-patient relationship
- Explanation about asthma and factors that influence it
- Starting appropriate medication
- Training about correct inhalation technique
- Reinforcement on importance of child's adherence to medication and avoidance to trigger factors
- Written Asthma Action Plan (WAAP)
- Follow up appointment in 4-12 weeks

Follow-Up Visit

1. Asthma Control review
The interval of reassessment can vary from once in 2 weeks (for a patient with poor control undergoing adjustment in medication) to once in 6 months for a patient with very well controlled condition.
2. Ask for "SPICE"
 - Symptoms
 - Parental concerns

- Inhaler techniques
 - Compliance / Adherence
 - Environmental triggers avoidance
3. Symptom control assessment with GINA symptoms control tool or Asthma Control Test
 4. Review growth chart

Assessment of Asthma Control

GINA Symptom Control Tool

GINA Assessment of asthma symptom control in Children 5 years and younger

In the past 4 weeks, has the child had:

Daytime asthma symptoms for more than a few minutes, more than once a week?	Yes/No
Any activity limitation due to asthma? (Runs / play less than other children, tires easily during walks / playing)	Yes/No
Reliever medication needed more than once a week? (exclude reliever before exercise)	Yes/No
Any night waking or night coughing due to asthma?	Yes/No

Well controlled : None of these
Partly controlled : 1-2 of these
Uncontrolled : 3-4 of these

GINA Assessment of asthma symptom control in Children 6-11 years

In the past 4 weeks, has the child had:

Daytime asthma symptoms for more than twice a week?	Yes/No
Any activity limitation due to asthma?	Yes/No
Reliever medication needed more than twice a week? (exclude reliever before exercise)	Yes/No
Any night waking due to asthma?	Yes/No

Asthma Control Test (ACT®)

This is a 7-item, patient and carer-administered questionnaire for assessing asthma control in children aged 4-11 years old.

Have your Paediatric patient answer the following questions:

1. How is your asthma today?				SCORE
 0 Very bad	 1 Bad	 2 Good	 3 Very good	
2. How much of a problem is your asthma when you run, exercise or play sports?				
 0 It's a big problem, I can't do what I want to do.	 1 It's a problem and I don't like it.	 2 It's a little problem but it's okay.	 3 It's not a problem.	
3. Do you cough because of your asthma?				
 0 Yes, all of the time.	 1 Yes, most of the time.	 2 Yes, some of the time.	 3 No, none of the time.	
4. Do you wake up during the night because of your asthma?				
 0 Yes, all of the time.	 1 Yes, most of the time.	 2 Yes, some of the time.	 3 No, none of the time.	

Ask the parents to answer the following questions:

5. During the <u>last 4 weeks</u> , on average, how many <u>days per month</u> did your child have any daytime asthma symptoms?						
5 Not at all	4 1-3 days/mo	3 4-10 days/mo	2 11-18 days/mo	1 19-24 days/mo	0 Everyday	
6. During the <u>last 4 weeks</u> , on average, how many <u>days per month</u> did your child wheeze during the day because of asthma?						
5 Not at all	4 1-3 days/mo	3 4-10 days/mo	2 11-18 days/mo	1 19-24 days/mo	0 Everyday	
7. During the <u>last 4 weeks</u> , on average, how many <u>days per month</u> did your child wake up during the night because of asthma?						
5 Not at all	4 1-3 days/mo	3 4-10 days/mo	2 11-18 days/mo	1 19-24 days/mo	0 Everyday	TOTAL

A score of 19 or less suggests poor asthma control.

A score of 20 or above suggests the child's asthma may be under control.

A score of 27 indicates 'Total control' of asthma.

Asthma Control Test for Children Aged 4 - 11 Years Old ©

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This is a 5-item, patient-administered questionnaire for assessing asthma control in children aged 12 and above.

Asthma Control Test (ACT)

- In the past 4 weeks, how much of the time did your asthma keep you from getting as much done at work, school or at home? Score

All of the time	1	Most of the time	2	Some of the time	3	A little of the time	4	None of the time	5
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- During the past 4 weeks, how often have you had shortness of breath?

More than once a day	1	Once a day	2	3 to 6 times a week	3	Once or twice a week	4	Not at all	5
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- During the past 4 weeks, how often did your asthma symptoms (wheezing, coughing, shortness of breath, chest tightness or pain) wake you up at night or earlier than usual in the morning?

4 or more nights a week	1	2 or 3 nights a week	2	Once a week	3	Once or twice	4	Not at all	5
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- During the past 4 weeks, how often have you used your rescue inhaler or nebulizer medication (such as albuterol or salbutamol)?

3 or more times per day	1	1 or 2 times per day	2	2 or 3 times per week	3	Once a week or less	4	Not at all	5
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- How would you rate your asthma control during the past 4 weeks?

Not controlled at all	1	Poorly controlled	2	Somewhat controlled	3	Well controlled	4	Completely controlled	5
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Patient Total Score

Based on a five-point scoring system,
 A score of 19 or less implies poor asthma control.
 A score of 20 -24 indicates 'well controlled' asthma
 A maximum score of 25 indicates 'total control' of asthma.

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Titrating Inhaled Corticosteroid

If child is well controlled and maintained for at least 3 months:

- Consider tapering inhaled corticosteroid treatment gradually to lowest effective dose.
- Children with high risk of poor asthma outcomes (e.g. previous ICU admission / intubation) and children with previous relapse after tapering of ICS dose, should be tapered cautiously.
- If ICS is tapered down or stopped, schedule follow-up in 3-6 weeks to review symptoms.

If child is partly controlled or uncontrolled, check the following:

- Verify diagnosis
- Assess inhaler technique
- Check adherence to medication and avoidance of trigger factors
- Management of co-morbid conditions like allergic rhinitis, GERD, etc.
- Review medication dose

Most of the children will respond to first line of low dose ICS if above factors are corrected. If all the above factors have been corrected, consider stepping up treatment.

Assessment of Risk Factors for Poor Asthma Outcomes

- Assess risk factors at diagnosis and periodically, particularly for patients experiencing exacerbations.
- For children 6-11 years and adolescents, measure lung function at diagnosis / start of treatment, 3-6 months after starting controller treatment, then periodically (e.g. at least once every 1-2 years, but more often in at-risk patients and those with severe asthma).

Features of patients at increased risk of adverse events in future:

(Having any of these risk factors increases the patient's risk of exacerbation even if they have few asthma symptoms)

- History of severe asthma exacerbations requiring intubation or care in the High Dependency or Intensive Care Unit
- ≥ 1 severe exacerbation in last 12 months
- High SABA use [> 1 canister of SABA (e.g. salbutamol) per month]
- Inadequate ICS, poor adherence with controller medications, or incorrect inhaler technique
- Comorbidities: Obesity, chronic rhino-sinusitis, GERD, confirmed food allergy
- Exposures: smoking, air pollution, allergens (e.g. house dust mites, cockroach, pets, mould)
- Major psychological or socioeconomic problems for child or family
- Low initial FEV₁, high BD reversibility
- Blood eosinophilia

Risk factors for persistent airflow limitation:

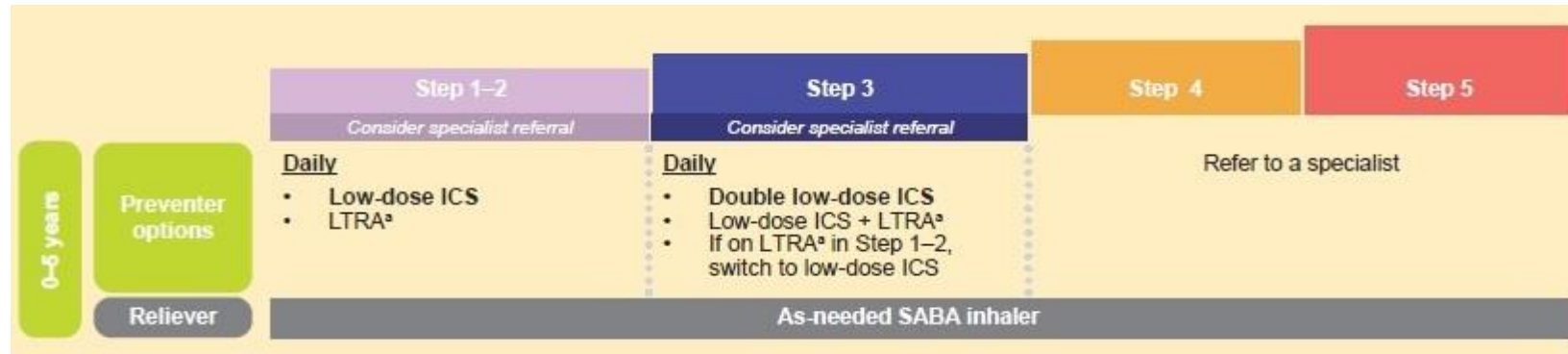
- Severe asthma with several hospitalizations
- History of bronchiolitis in the first 3 months of age
- History of maternal smoking in pregnancy, preterm birth, low birth weight and neonatal ventilation.

Risk factors for medication side-effects:

- Systemic: Chronic use of moderate to high dose of ICS may reduce growth velocity in pre-pubertal children and slight reduction in adult final height.⁹ However, poorly controlled asthma itself may have much greater impact on child's growth.
- Local: With good inhaler technique of using spacers, local side-effects are not common in children. When ICS is used without spacer, local side effects such as oral thrush should be looked for.

Pharmacological Treatment

Asthma Treatment Steps for Children Aged 5 Years and Younger¹⁰



The patient's current treatment and level of control determine the selection of pharmacologic treatment. If asthma is not controlled on the current treatment, treatment should be stepped up until control is achieved. Control is usually maintained for at least 3 months before an attempt is made to step down the treatment, with the aim to establish the lowest step and dose of treatment that maintains control.

In children aged 0 to 5 years, **long-term treatment with SABA alone (without preventer) for asthma could be used only if the child fulfils all the following criteria:**

- No history of ICU admission or intubation for asthma
- No more than 3 exacerbations over the past year
- Normal lung function test over the past year (if available)
- No night awakening due to asthma over the past 4 weeks
- No exercise limitations due to asthma over the past 4 weeks
- Asthma symptoms no more than once over the past 4 weeks
- SABA used no more than once over the past 4 weeks

Step 1-2: Low-dose ICS, plus as-needed inhaled SABA. Consider specialist referral.

- Oral bronchodilator therapy is not recommended due to its slower onset of action and higher rate of side-effects
- It is important to discuss decision to prescribe controller treatment and the choice of treatment with child's parents or carer, including relative benefits and risk of treatments.
- Other option: Daily LTRA or intermittent short course of ICS at onset of respiratory distress.
***Blackbox Warning for Montelukast:** Healthcare providers are advised to consider the benefit of treatment and risk of neuropsychiatric effects before prescribing montelukast. Parents should be counselled about risk of impact on sleep, behaviour and mood changes (including serious mental side effects such as suicidal thoughts or action, and depression).

Step 3: Double low-dose ICS, plus as-needed SABA. Consider specialist referral

- There are insufficient data about the efficacy and safety of ICS-LABA in children < 4 years old and it is not approved for use in this age group

Step 4 & 5: Continue controller treatment and refer to a specialist

Asthma Treatment Steps for Children Aged 6 to 11 Years¹⁰

Age	Treatment category	Step 1–2	Step 3	Step 4 (consider specialist referral)	Step 5 (refer to a specialist)
6–11 years	Preventer options	<u>Daily</u> - Low-dose ICS - LTRA^a	<u>Daily</u> - Low-dose ICS-LABA - Low-dose ICS + LTRA^a - Medium-dose ICS	<u>Daily</u> - Medium-dose ICS-LABA - Medium-dose ICS + LTRA^a	Continue treatment as per Step 4 and consider add-on treatment with biologic agents for asthma ^a , or low-dose OCS
	Reliever	<u>As needed</u> Low-dose ICS whenever SABA is used^c	<u>MART^f</u> Daily low-dose ICS-formoterol + as-needed low-dose ICS-formoterol	<u>Possible adjustments to daily preventer options above:</u> - Add LTRA^a to medium-dose ICS-LABA - Increase to high-dose ICS-LABA or high-dose ICS + LTRA^a - Add tiotropium	
As-needed SABA inhaler (as-needed ICS-formoterol is the reliever if MART ^f is chosen in Step 3)					

The patient's current treatment and level of control determine the selection of pharmacologic treatment. If asthma is not controlled on the current treatment, treatment should be stepped up until control is achieved. Control is usually maintained for at least 3 months before an attempt is made to step down the treatment, with the aim to establish the lowest step and dose of treatment that maintains control.

Step 1-2: Daily low dose ICS, plus as-needed SABA

- Other options: taking ICS whenever SABA is taken – evidence for this is still limited in this age group
- Other option: Daily LTRA with as needed SABA; or taking low dose ICS whenever SABA is taken.

***Blackbox Warning for Montelukast:** Healthcare providers are advised to consider the benefit of treatment and risk of neuropsychiatric effects before prescribing montelukast. Parents should be counselled about risk of impact on sleep, behaviour and mood changes (including serious mental side effects such as suicidal thoughts or action, and depression).

Step 3: Low dose ICS-LABA, plus as needed SABA; or medium dose ICS plus as needed SABA

- Other option: Low dose ICS with daily LTRA, with as needed SABA.

***Blackbox Warning for Montelukast:** Healthcare providers are advised to consider the benefit of treatment and risk of neuropsychiatric effects before prescribing montelukast. Parents should be counselled about risk of impact on sleep, behaviour and mood changes (including serious mental side effects such as suicidal thoughts or action, and depression).

- Other option: MART: Daily low-dose ICS-formoterol plus as needed low-dose ICS-formoterol

Step 4: Medium dose ICS-LABA, plus as needed SABA. Consider specialist referral.

- Patients who do not achieve good asthma control despite Step 4 levels of treatment may have refractory asthma and should be reviewed by a specialist.

Step 5: Refer specialist for phenotypic assessment and consideration of add-on treatment. Management should be supervised directly by specialist.

Low, Medium, High Daily Doses of Inhaled Corticosteroids

Children Aged 5 years and younger²

Drug	Low Total Daily Dose (mcg)
Beclomethasone Dipropionate (pMDI, extrafine particle, HFA)	50 (ages 5 years and older)
Fluticasone Propionate (pMDI, standard particle, HFA)	50 (ages 4 years and older)

Children Aged 6 - 11 years ²

Drug	Low Daily Dose (mcg)	Medium Daily Dose (mcg)	High Daily Dose (mcg)
Beclomethasone Dipropionate (pMDI, extrafine particle*, HFA)	50 - 100	> 100 - 200	> 200
Budesonide (DPI)	100 - 200	> 200 - 400	> 400
Fluticasone Propionate (DPI)	50 - 100	> 100 - 200	> 200
Fluticasone Propionate (pMDI, standard particle, HFA)	50 - 100	> 100 - 200	> 200

DPI: Dry powder inhaler; HFA: Hydrofluoroalkane propellant

Combination of Inhaled Corticosteroids and Long Acting Beta Agonist

Drug	Dosage
Seretide 25/50® Evohaler (Fluticasone 50mcg/ Salmeterol 25mcg)	<u>For Children Aged 4 years and above:</u> Usual dose : 1 to 2 puffs once to twice daily Maximum dose : 2 puffs twice daily
Seretide 50/100® Accuhaler (Fluticasone 50mcg/ Salmeterol 50mcg)	<u>For Children Aged 4 years and above:</u> Usual dose : 1 puff once or twice daily Maximum dose : 1 puff twice daily
Symbicort® Rapihaler (Budesonide 80mcg/ Formoterol 2.25mcg)	<u>For Children Aged 6-11 years:</u> Recommended dose : 2 puff BD *When control of symptoms is achieved with twice daily regimen, tapering of treatment to ICS only can be offered. If long-acting bronchodilator is required to maintain control, treatment can be titrated to lowest effective dose of once daily.
Symbicort® Turbuhaler (Budesonide 80mcg/ Formoterol 4.5mg) <i>*Not available in NUP Formulary</i>	<u>For Children Aged 6-11 years:</u> Recommended dose : 2 puff BD *When control of symptoms is achieved with twice daily regimen, tapering of treatment to ICS only can be offered. If long-acting bronchodilator is required to maintain control, treatment can be titrated to lowest effective dose of once daily.

Note: LABAs should NOT be used without concomitant inhaled corticosteroids in asthma

Recommended Inhaler Devices for Children:

A pressurized metered-dose inhaler (pMDI) with a valved spacer (with or without a face mask, depending on child's age) is the preferred delivery system for children of all age group.

< 4 years pMDI plus spacer with face mask
 ≥ 4 years pMDI plus spacer with mouthpiece

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of asthma, including health promotion and prevention / treatment of complications
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care

2. Care Manager

- Evaluate understanding and provide education on asthma, good asthma control and lifestyle measures
- Assess and identify reasons for suboptimal / poor adherence to treatment plans and devise strategies to motivate the patients
- Provide education on preventer and reliever inhalers, inhaler technique and use of asthma action plan

3. Pharmacist

- Assess and teach use of various inhalers and delivery devices

IV. SPECIAL SITUATIONS

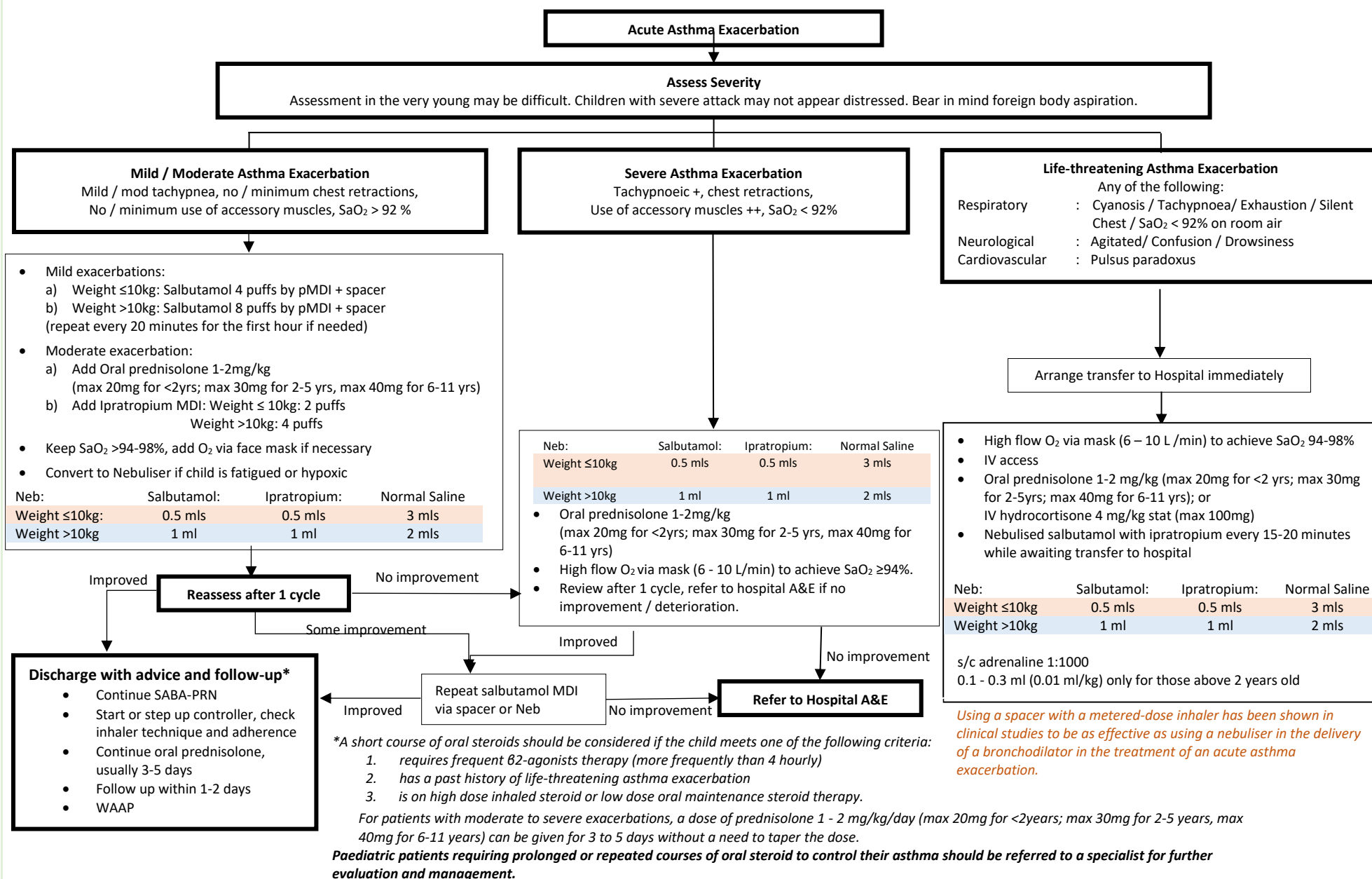
Asthma Exacerbations

1. How Severe is the Asthma Attack?

SEVERITY SYMPTOMS	MILD	MODERATE	SEVERE	RESPIRATORY ARREST IMMINENT
Breathlessness	While walking Can lie down	While at rest (infant – softer, shorter cry) Prefer sitting	While at rest Hunched forward	
Feeding (infant)	Feeds normally	Difficulty feeding	Stops feeding	
Talks in...	Sentences	Phrases	Words	
Alertness	May be agitated	Usually agitated	Usually agitated	Drowsy or confused
SEVERITY SIGNS	MILD	MODERATE	SEVERE	RESPIRATORY ARREST IMMINENT
Respiratory rate*	Increased	Increased	Increased	
Use of accessory muscles / suprasternal retractions	Usually not	Usually	Usually	Paradoxical thoraco– abdominal movement
Central cyanosis	Absent	Absent	May to be present	
Wheeze	Moderate, often only end expiratory	Loud	Chest may be quiet	Absence of wheeze
Pulse rate*	<100 beats/minute	Increased	>180 beats/minute (0-3 years) >150 beats/minute (4-5 years)	Bradycardia
FUNCTIONAL ASSESSMENT				
Pulse Oximetry PaO ₂ (on air)	> 92%	> 92%	< 92%	

* The presence of several parameters, but not necessarily all, indicate the general classification of the attacks. Also note any prior usage of bronchodilator as this may alter the presenting clinical pictures. Please refer to the table on BA-14 for the guide to normal respiratory and pulse rates in infants and children.

2. Management of Acute Asthma Exacerbation in Children



Guide to Normal Respiratory and Pulse Rates

AGE	RESP RATE (MIN)
< 2 months	< 60
2 - 12 months	< 50
1 - 5 years	< 40
6 - 8 years	< 30

AGE	PULSE RATE (MIN)
2 - 12 months	< 160
1 - 2 years	< 120
2 - 8 years	< 110

V. REFERRALS

Presence of Risk Factors for Death from Asthma

1. Prior intubation and mechanical ventilation for asthma
2. Hospitalisation or emergency care visit for asthma in the past year
3. Current use of systemic corticosteroids or recent withdrawal from systemic corticosteroids
4. Not currently using inhaled corticosteroids
5. Use of > 1 canister of inhaled short-acting β 2-agonist within 1 month
6. History of psychiatric disease or psychosocial problems

Patients at high risk of dying with asthma require special attention, monitoring and care, particularly intensive education, including advice to seek medical care early during an exacerbation.

When Diagnosis is Unclear

1. Atypical signs and symptoms.
2. Other conditions complicate asthma or its diagnosis, e.g. heart failure, unsure of diagnosis
3. Additional diagnostic testing is indicated.

Indications for Referral to the Paediatric Specialist for Further Evaluation and Management

1. Patients with high risk asthma with poor control.
2. Patient aged 5 years old and younger (Referral to specialist should be considered for this age group).
3. Patients who remain symptomatic, show suboptimal response to therapy.
4. Patients requiring high doses of inhaled steroids (Beclomethasone Dipropionate or Budesonide ≥ 400 mcg/day).

VI. RECOMMENDED CARE COMPONENTS FOR ASTHMA

Recommended Care Components	Minimum Frequency	Remarks
Asthma Control Assessment [^] [Global Initiative for Asthma (GINA) Score, Asthma Control Test*]	At least twice a year	
Smoking Assessment [^]	Annually for smokers; Once-off for non-smokers, unless there is a change in smoking habit	Assessment on smoking habits (estimated sticks/day; zero for non- or ex-smoker) and provide smoking cessation counselling
Written Asthma Action Plan	Upon diagnosis, recommended annually	
Spirometry	Recommended at or soon after diagnosis, or when clinically indicated (if age appropriate)	
Influenza Vaccination and Pneumococcal Vaccination	As recommended under the National Childhood Immunisation Schedule	

[^] These are reportable clinical indicators

* Asthma Control Test is not used for asthma control assessment in NUP

VII. ABBREVIATIONS

<	Less than
≤	Less than or equal to
>	More than
≥	More than or equal to
BDP	Beclomethasone Dipropionate
BUD	Budesonide
Ca	Carcinoma
COPD	Chronic Obstructive Pulmonary Disease
CXR	Chest X-Ray
ETT	Endotracheal tube
FEV1	Forced expiratory volume in 1 second
i/v	Intravenous
ICS	Inhaled corticosteroids
ICU	Intensive care unit
LABA	Long-acting beta ₂ agonist
LOW	Loss of weight
LTRA	Leukotriene receptor antagonist
MART	Maintenance and reliever therapy
mths	Month(s)
O ₂	Oxygen
PEF	Peak expiratory flow
pMDI	Pressurised Metered Dose Inhaler
s/c	Subcutaneous
SABA	Short-acting beta ₂ agonist
TCU	To-see-you (i.e. review appointment)
Yr	Year(s)

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